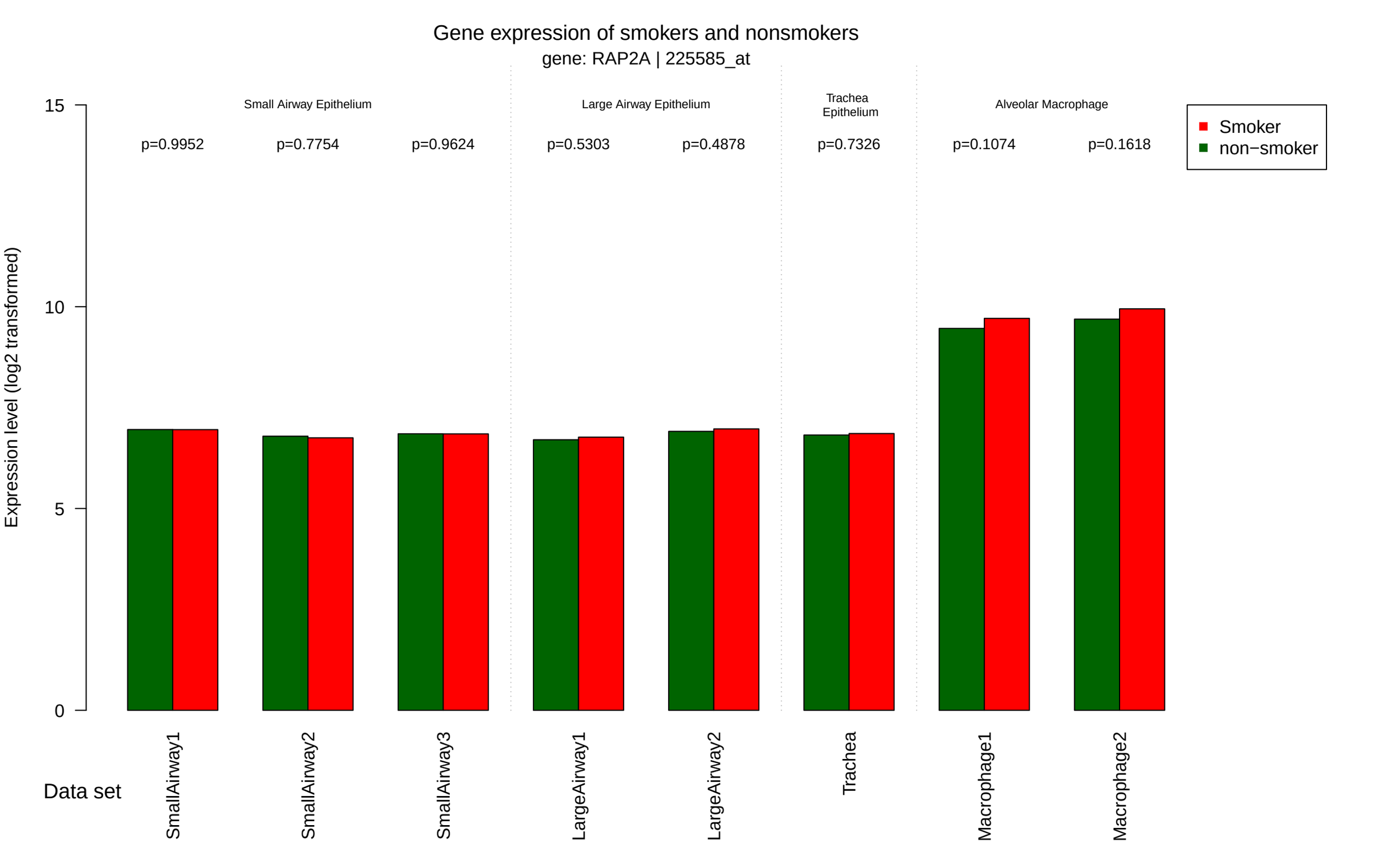
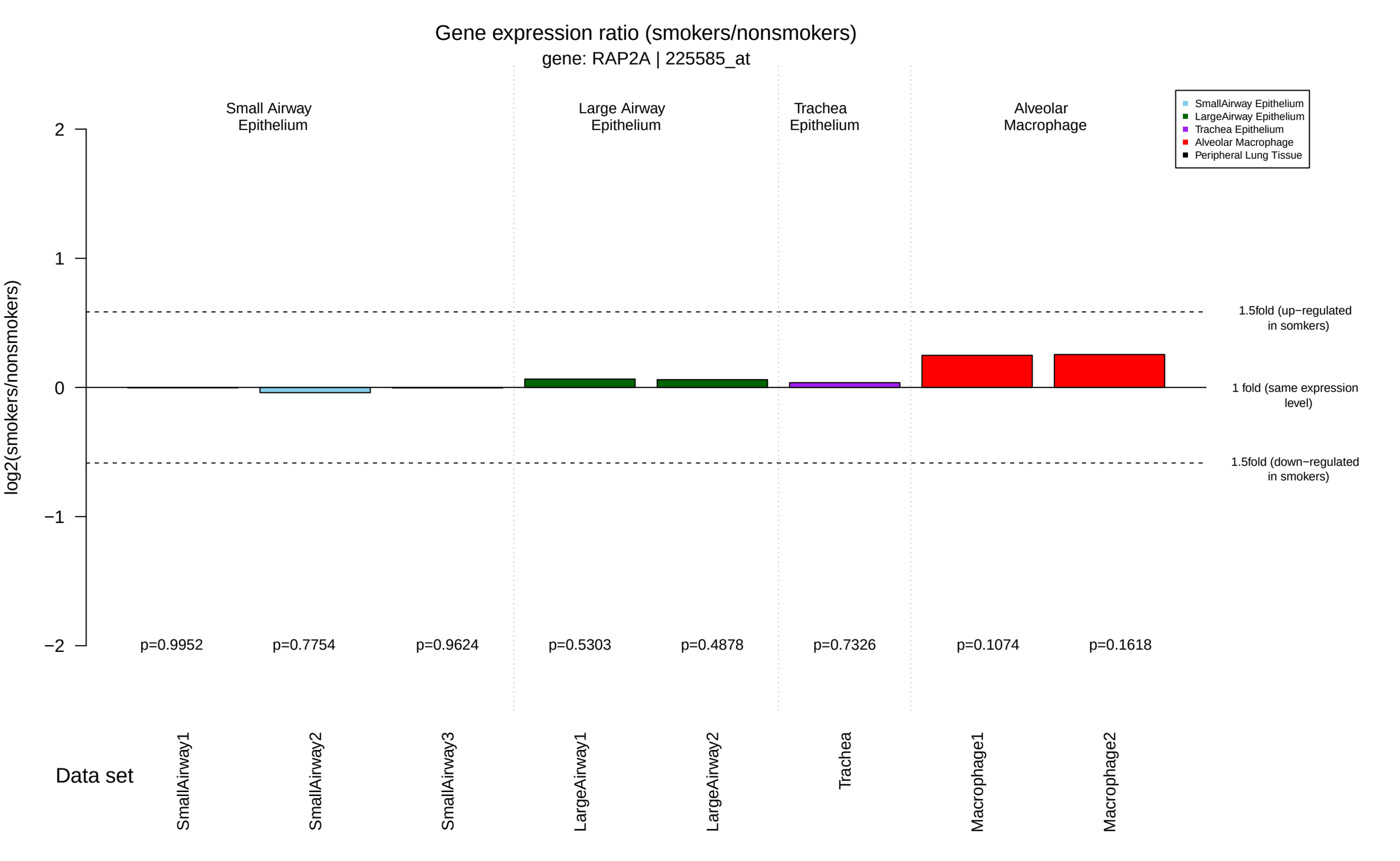
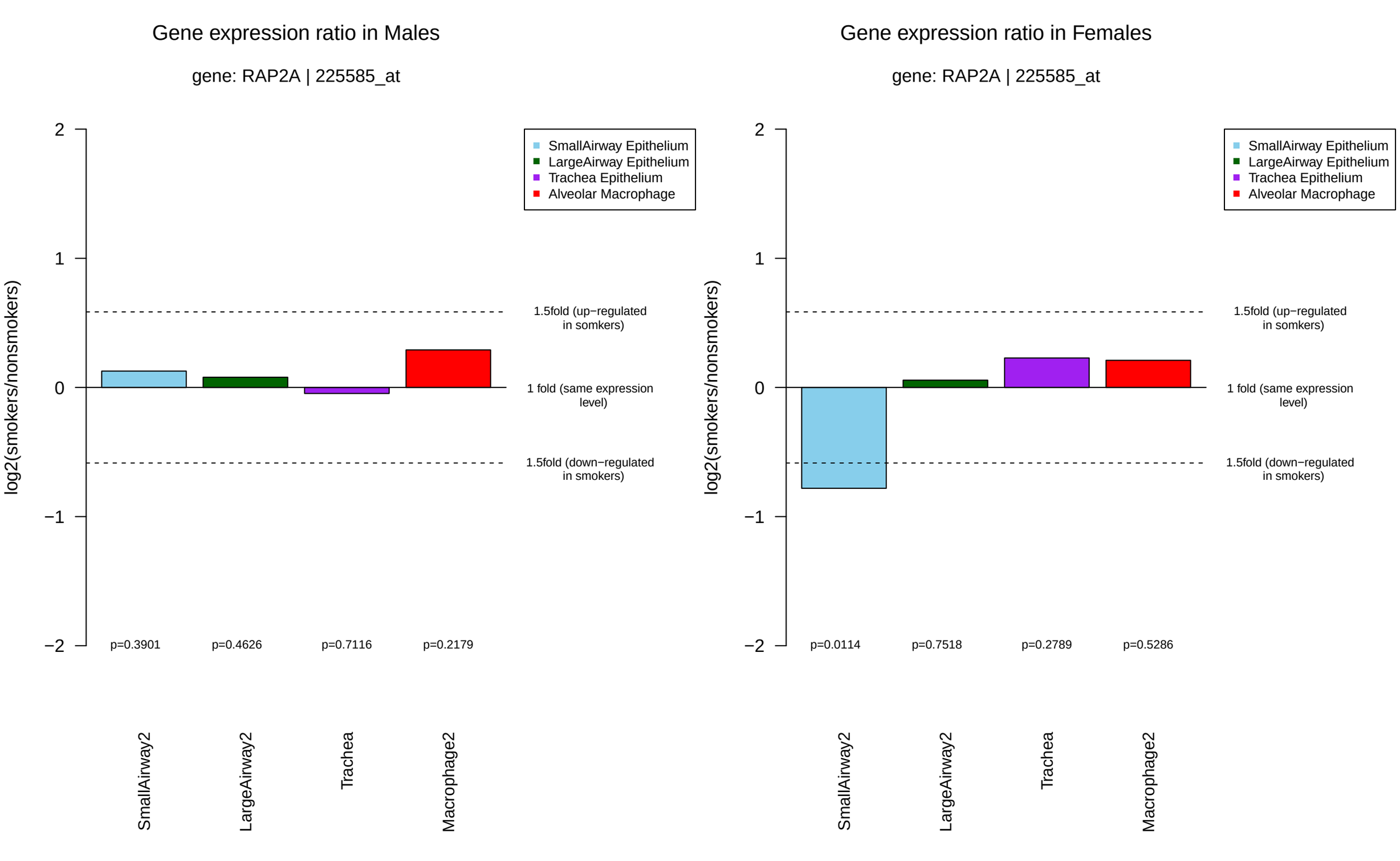
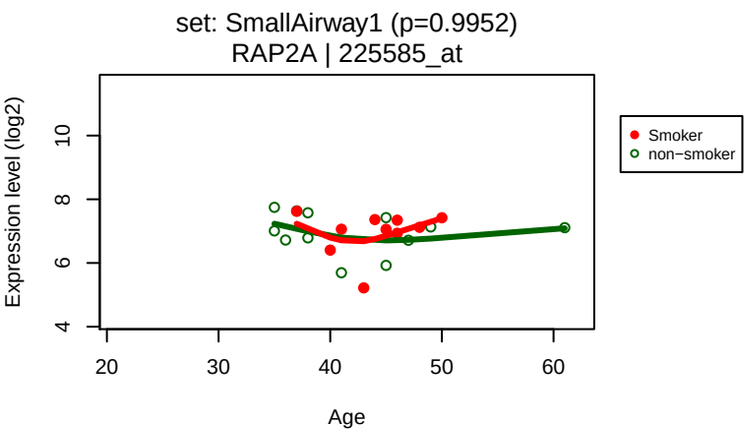
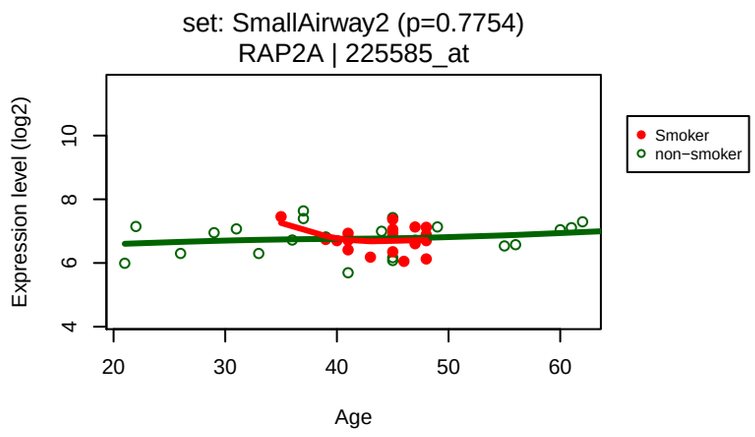
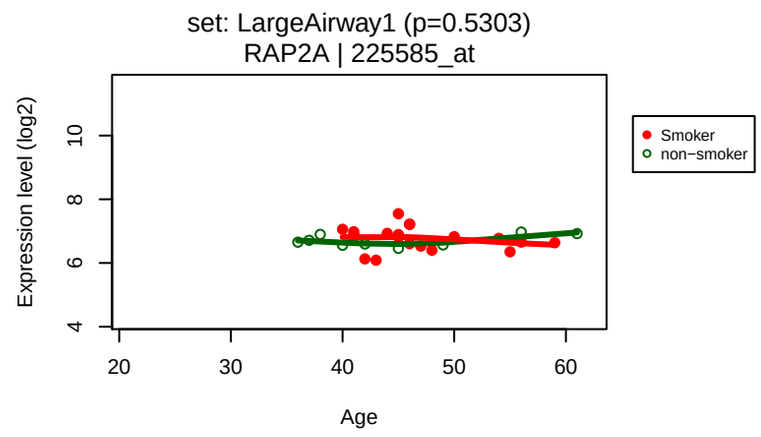

Gene expression of smokers and nonsmokers
gene: RAP2A | 225585_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9952
p=0.7754
p=0.9624
p=0.5303
p=0.4878
p=0.7326
p=0.1074
p=0.1618
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

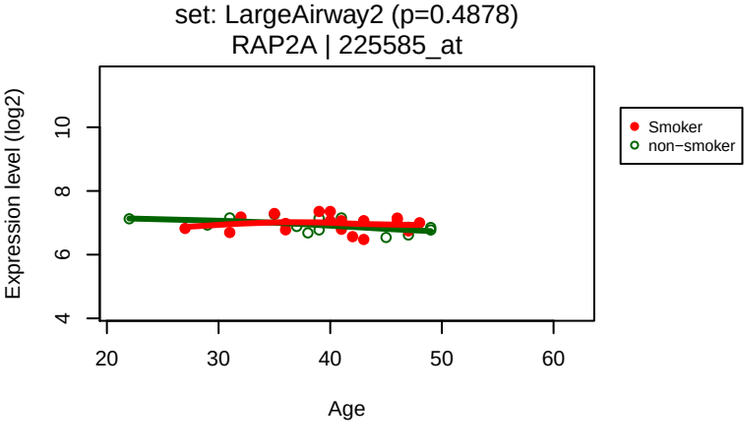

Gene expression ratio (smokers/nonsmokers)
gene: RAP2A | 225585_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9952
p=0.7754
p=0.9624
p=0.5303
p=0.4878
p=0.7326
p=0.1074
p=0.1618
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

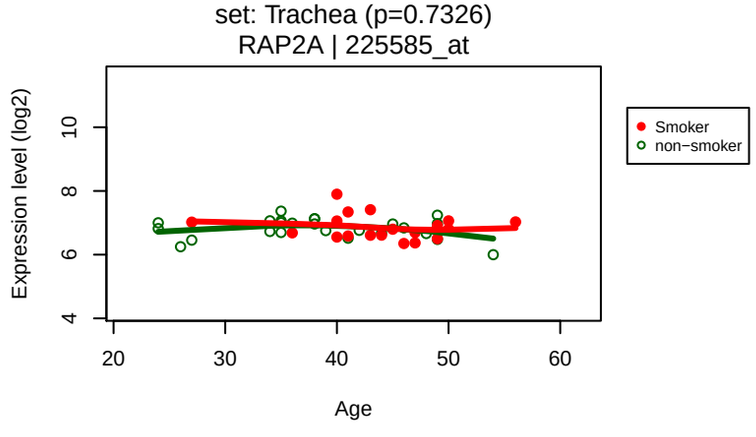

Gene expression ratio in Males
gene: RAP2A | 225585_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3901
p=0.4626
p=0.7116
p=0.2179
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAP2A | 225585_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0114
p=0.7518
p=0.2789
p=0.5286
SmallAirway2
LargeAirway2
Trachea
Macrophage2

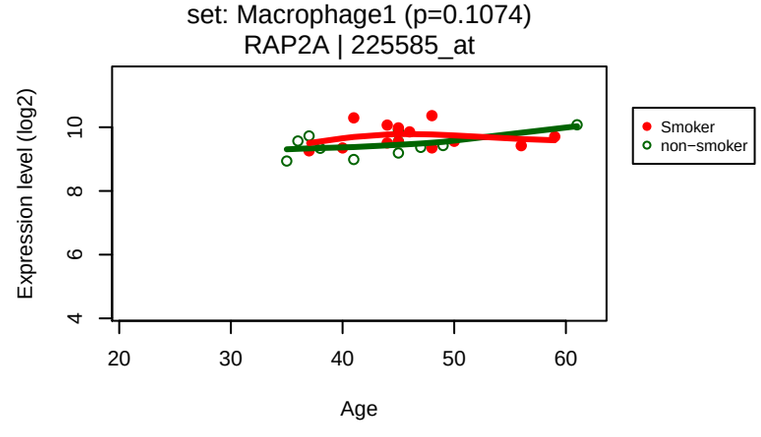

set: SmallAirway1 (p=0.9952)
RAP2A | 225585_at
Expression level (log2)
Age
Smoker
non−smoker

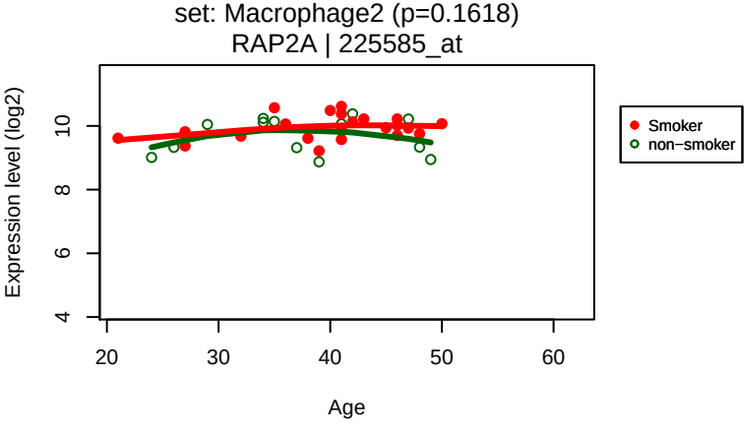

set: SmallAirway2 (p=0.7754)
RAP2A | 225585_at
Expression level (log2)
Age
Smoker
non−smoker

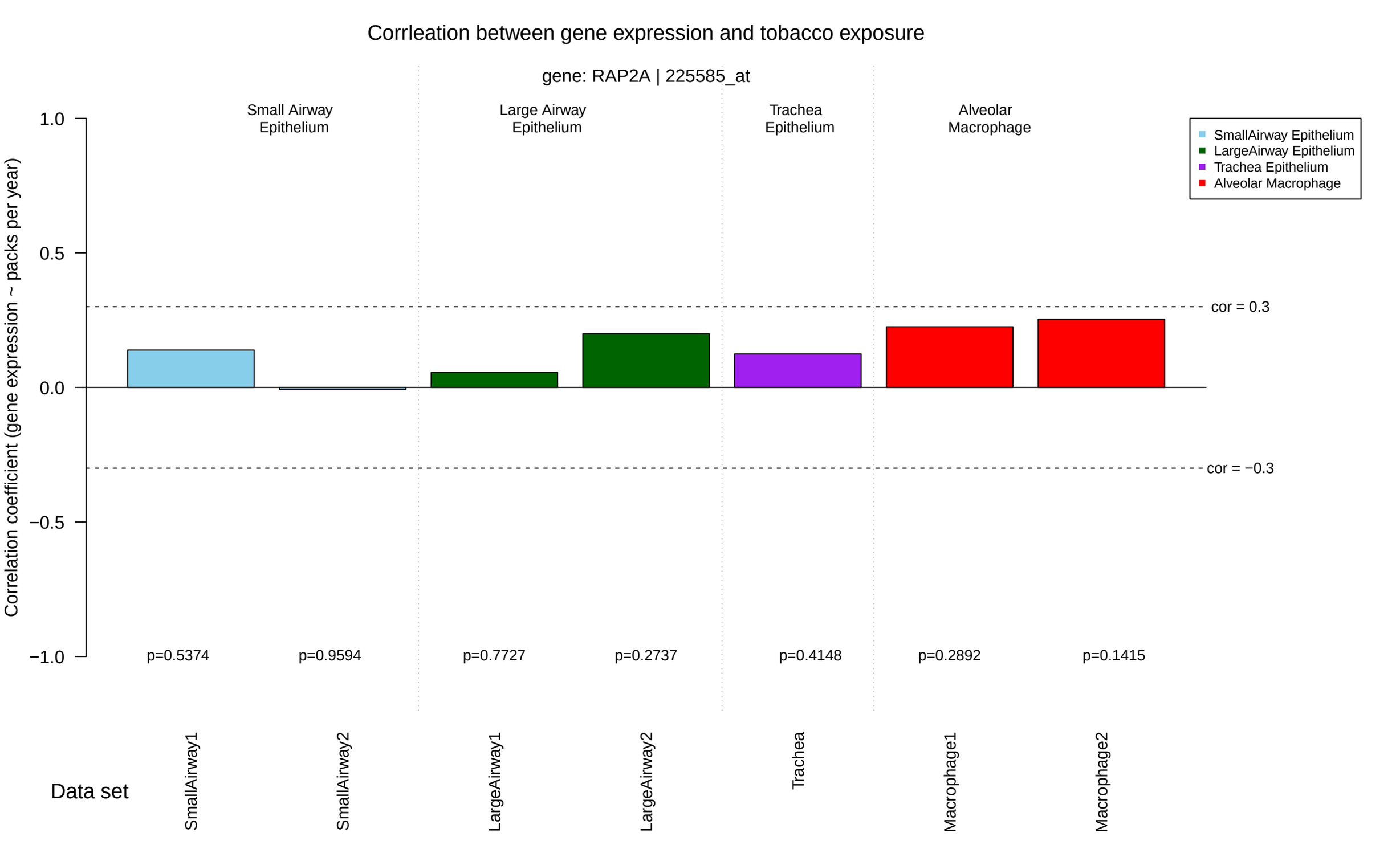

set: LargeAirway1 (p=0.5303)
RAP2A | 225585_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4878)
RAP2A | 225585_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7326)
RAP2A | 225585_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1074)
RAP2A | 225585_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1618)
RAP2A | 225585_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAP2A | 225585_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5374
p=0.9594
p=0.7727
p=0.2737
p=0.4148
p=0.2892
p=0.1415
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2